SOLAR▸SWARM

Bundle

Martin Pham | Perth, AU

Solar Swarm — Business Overview

Owner: Martin Pham | Perth, Australia | AI Automation Consultant

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Executive Summary

Solar Swarm is an AI-powered sales automation platform purpose-built for Australian solar installation companies. It eliminates the two biggest revenue leaks in solar sales: slow lead response times and inconsistent follow-up.

The platform integrates directly into a client's existing GoHighLevel CRM and begins operating within 48 hours of setup. Every inbound lead is automatically scored, called by an AI voice agent, and moved through the pipeline — without any action required from the sales team.

Revenue model: $1,500–2,000 AUD/month retainer per client

Margin: 80–90% (total operating costs ~$400 AUD/month)

Break-even: 1 client

The Problem We Solve

Australian solar SMEs (5–15 salespeople) share a universal pattern:

PAIN POINT	BUSINESS IMPACT
Leads sit uncontacted for 2–4 hours	Industry data: response after 5 min = 9x more conversions
Sales reps cherry-pick leads	Hot prospects go cold while unqualified ones get called
No consistent follow-up system	80% of solar deals close after 3+ touchpoints; most reps stop at 1
Owner has no visibility	Can't see pipeline health, call volumes, or what's working
Staff turnover disrupts process	Every new hire needs retraining on the follow-up playbook

Solar Swarm replaces all five failure points with automated, AI-driven systems that run 24/7.

What the Platform Delivers

Lead Qualification (Immediate)

Every new enquiry receives a GPT-4o qualification score from 1–10 within seconds of arriving in GHL. The score is based on:

- Solar interest signals (roof ownership, electricity bill, timeline)
- Contact completeness (phone, email, suburb)
- Competitive context (existing quotes, urgency)

AI Voice Outreach (Within Minutes)

Hot leads (score ≥ 7) trigger an outbound AI phone call via Retell AI. The AI:

- Speaks in natural Australian English
- Qualifies the lead further using a customised script
- Books a site assessment into the sales calendar
- Updates the GHL pipeline with call outcome

Pipeline Automation

- Contacts tagged and moved through stages automatically
- Slack alerts sent to owner and rep for hot leads
- Cold leads placed into long-term nurture sequences
- Weekly performance report delivered every Monday

Live Dashboard

A web-based dashboard (port 5003) shows:

- Pipeline stage counts in real time
- Lead qualification scores and call outcomes
- Swarm experiment status and AI activity
- Budget utilisation and circuit breaker state

Target Client Profile

ATTRIBUTE	SPECIFICATION
Industry	Residential/commercial solar installation
Location	Australia (primarily WA, QLD, VIC, NSW)
Company size	5–15 salespeople
Monthly lead volume	50–500 enquiries
CRM	GoHighLevel (or willing to adopt)
Current pain	Slow response, inconsistent follow-up, no visibility
Decision maker	Owner/Director
Budget signal	Already paying for GHL ($97–$297/month) — understands SaaS

Competitive Differentiation

FEATURE	SOLAR SWARM	GENERIC CRM AUTOMATION	OFFSHORE VA	IN-HOUSE SDR
Response time	< 2 minutes	Hours (if configured)	Business hours	Business hours
24/7 operation	Yes	Yes	No	No
Industry-specific scoring	Yes	No	No	Trained only
Voice AI calls	Yes	No	No	Yes
Setup time	48 hours	Weeks	Days	Months
Monthly cost	$1,500–2,000	$200–500	$3,000–5,000	$8,000–12,000
Scales without cost	Yes	Partly	No	No

Revenue Model

Retainer Structure

```
Tier 1 — Foundation:  $1,500/month
  ├── Lead qualification + pipeline automation
  ├── Weekly performance reports
  └── Standard Slack alerts

Tier 2 — Growth:      $2,000/month
  ├── Everything in Tier 1
  ├── AI voice outreach (Retell integration)
  ├── Custom qualification criteria
  └── Monthly strategy review call

Add-ons (future):
  ├── Multi-location support:     +$500/location
  ├── Custom AI persona/voice:    +$300 one-time
  └── Advanced A/B experiment:    $500 per experiment
```

Unit Economics

```
Revenue per client:          $1,750/month (average)
OpenAI API costs:            ~$150/month (GPT-4o at scale)
Retell AI voice costs:       ~$100/month (500 calls × $0.20)
Infrastructure/hosting:      ~$50/month
Slack/tools:                 ~$50/month
Gross cost per client:       ~$350–400/month
Gross margin:                80%+

At 5 clients:
  Revenue:    $8,750/month
  Costs:      ~$2,000/month
  Profit:     ~$6,750/month
```

Growth Roadmap

Phase 1 — Proof of Concept (Current)

- First 1–2 clients onboarded at reduced rate ($1,000/month)
- Validate qualification accuracy and AI call quality
- Build case studies with real metrics

Phase 2 — Productise (Months 3–6)

- Standard onboarding playbook (48-hour setup)
- Self-service dashboard for clients
- Referral incentive programme
- Target: 5 clients

Phase 3 — Scale (Months 6–12)

- Agency partner model (other consultants white-label)

- Multi-industry expansion (HVAC, pools, property)
- Hiring first contractor support
- Target: 15+ clients, $25,000+/month revenue

Risk Register

RISK	PROBABILITY	IMPACT	MITIGATION
Client churns after month 1	Medium	High	Strong onboarding + first-week WOW moments
OpenAI API price increase	Low	Medium	Margin buffer; switch to Anthropic if needed
GHL changes API	Low	High	CRM router abstraction; HubSpot/SF fallback
AI call quality complaints	Medium	High	Human review toggle; call recording review
Competitor copies approach	High	Medium	Relationships + speed advantage
Regulatory (AI calling laws)	Low	High	Human handoff option; full disclosure

Success Metrics

METRIC	TARGET
Lead response time	< 5 minutes for hot leads
Qualification accuracy	> 80% agreement with human assessment
AI call answer rate	> 40% of outbound calls answered
Pipeline velocity	20%+ increase in stage progression speed
Client retention	> 90% at 3 months
NPS score	> 50

Solar Swarm — Sales Playbook

For: Martin Pham (and future sales team)

Purpose: Outreach, qualification, and closing process for solar SME clients

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Ideal Client Profile (ICP) Quick Reference

YES signals:

- Solar installation company, 5–15 salespeople
- Already using GoHighLevel (or open to it)
- Owner-operated or owner-managed
- Mentions "leads going cold", "can't follow up fast enough", "missing jobs"
- Google reviews mention fast response or complain about slow contact
- Running Google Ads or Facebook Ads (spending on leads = lead pain)

NO signals (for now):

- Fewer than 3 salespeople (too small to justify cost)
- Enterprise (50+ staff — different buying process)
- Not in Australia
- Just getting started (no lead volume yet)

Outreach Sequence

Day 1 — LinkedIn Connection + Note

```
Subject: [First Name] – quick question about your lead follow-up

Hey [First Name],

Came across [Company Name] – looks like you're running a solid operation in [City].

Quick question: how long does it typically take your team to follow up a new enquiry?

Most solar companies I speak to are losing jobs because leads go cold before anyone calls.
We've built an AI system that calls every hot lead within 2 minutes, 24/7.

Worth a 15-minute chat?

Martin
```

Day 3 — Follow-Up (If No Response)

```
Hey [First Name],

Following up on my note about lead response times.

One of our clients in [nearby city/state] was losing roughly 3–4 jobs a week to competitors
who just called first. The AI now handles every new enquiry within minutes — they haven't
changed anything else.

Happy to show you a quick demo. 15 minutes on Zoom?

Martin
```

Day 7 — Value-Add (Case Study Angle)

```
[First Name],

Sharing something that might be useful regardless.

We tracked response time data across 12 solar companies — the ones responding within 5 minutes
closed 31% more jobs than those responding after an hour. Same ad spend, same leads.

If that gap exists in your business, I can show you exactly how we close it.

Worth a quick call?

Martin
```

Day 14 — Soft Close / Park

```
[First Name],

Last note on this — I don't want to keep cluttering your inbox.

If now isn't the right time, totally understand. Solar season timing makes sense.

When things pick up, we should talk. In the meantime, here's what we do:
[Link to one-pager or deck]

I'll check back in a few months.

Martin
```

Discovery Call Framework (30 Minutes)

Opening (5 min)

"Before I show you anything, I want to understand your operation a bit. Okay if I ask a few questions?"

Key Discovery Questions

Volume and flow:

- "How many leads do you get per week across all channels?"
- "Where do most come from — Google, Facebook, referrals?"
- "What does a lead look like when it arrives in your system?"

Pain:

- "What happens to a lead the moment it comes in — who gets it, how fast do they call?"
- "Have you ever lost a job where you knew the lead was hot but just didn't get to them in time?"
- "What percentage of your leads would you say are actually worth calling?"

Current setup:

- "Are you on GoHighLevel? How are you using it?"
- "Do you have any automation set up — follow-up texts, emails, anything?"

Decision-making:

- "Is it just you who'd be making this decision, or is there someone else involved?"
- "If we showed you something that worked, what would you need to see to move forward?"

Pivot to Demo (5 min)

"Based on what you've told me, I think there are two or three things that are directly relevant to you. Can I show you?"

Demo Flow (15 min)

1. Show lead arriving in GHL (or describe the flow)
2. Show qualification score appearing
3. Show Slack alert firing
4. Play a sample AI call recording
5. Show dashboard with pipeline health

Handling Objections (5 min)

See objection handler below.

Close (5 min)

"Based on everything you've seen — is this something that makes sense for your business right now?"

If yes: "Great — the way it works is a $1,500/month retainer. We can have you live in 48 hours. Want to lock in a start date?"

If "I need to think about it": "Totally fair. What's the main thing you want to think through?" → Address it, then re-ask.

Objection Handler

"It's too expensive."

"I hear you. Let me put it in context — if this system books you one extra job per month that you would have otherwise lost, what's your average job value?"

[They say $8,000–15,000.]

"So one job covers the retainer for six months. The question isn't really cost — it's whether the system works. And that's why we offer the first month as a trial. If it doesn't measurably improve your pipeline in 30 days, we'll refund it. Does that change the calculus?"

"We already have a CRM / we do follow-up ourselves."

"That's good — the system doesn't replace what you have, it speeds it up. It still goes into GHL, your team still closes the deals. The AI just makes sure no lead sits cold for more than a few minutes. Does your team currently call every lead within 5 minutes of arrival, even at 9pm on a Friday?"

"I'm not sure about AI calling customers."

"That's a fair concern. A couple of things: the AI only calls when a human hasn't — it's a backup, not a replacement. And the call recording goes into GHL so your team can listen and follow up. We can also set it to only call during business hours, or always hand off to a human after the opening. Would that make you more comfortable?"

"We're too busy right now."

"Busy is a good problem. Actually, that's exactly when this pays off most — when leads are coming in fast and your team is stretched, things fall through the cracks. This is designed for that scenario. Setup takes us 48 hours and we handle everything. Your team just keeps doing what they're doing."

"We've tried automation before and it didn't work."

"What happened? [Listen.] That's common — most automation tools are generic and need a lot of configuration. This is built specifically for solar companies in Australia. The qualification criteria, the call script, the pipeline stages — they're all set up for your industry out of the box."

Pricing and Packages

Quoting Guidelines

- Never start below $1,500/month (undermines positioning)
- Lead with $2,000 (Tier 2 with voice AI) — it's the better product
- Offer $1,500 only if they push back on budget
- First-month-free trial for qualified prospects only (not as default)
- Annual pre-pay: 10% discount (2 months free)

Proposal Format

Send a one-page proposal within 24 hours of a positive discovery call. Include:

1. Their specific pain (from the call)
2. What the system does for them
3. Investment (monthly, annual option)
4. Start date and onboarding steps
5. Call-to-action: "Reply YES to lock in [date]"

Post-Sale Handoff

Once signed:

1. Send onboarding questionnaire (see client-onboarding.md)
2. Schedule 60-min setup call within 48 hours
3. Configure GHL, set qualification criteria
4. Run first test lead through system
5. Send "go live" confirmation with dashboard access
6. Check in on Day 3, Day 7, Day 30

Key Metrics to Track

METRIC	WEEKLY TARGET
New LinkedIn connections	20
Outreach messages sent	40
Replies received	8+ (20% response rate)
Discovery calls booked	3
Demos delivered	2
Proposals sent	1
Closes	0.5 (1 per 2 weeks)

Solar Client Onboarding Guide

This guide is written for solar business owners. No technical knowledge required.

What You're Getting

Solar Swarm is an AI system that works in the background of your business, 24 hours a day, doing three things:

1. **Qualifies your leads** — Every time a new enquiry comes in, the AI scores them from 1 to 10 and decides whether to call them immediately, nurture them, or move on.
1. **Handles calls for you** — Hot leads (score 7 or higher with a phone number) get an outbound AI call automatically. The AI speaks in plain Australian English, asks the right questions, and books assessments into your calendar.
1. **Shows you what's working** — You get a weekly report and a live dashboard showing your lead pipeline, call outcomes, and what the AI is learning.

The system connects to your existing GoHighLevel (GHL) account. Nothing changes about how you use GHL — you just start seeing leads arrive pre-scored, pre-called, and pre-organised.

What We Need From You

To get you set up, we need:

1. Your GHL Sub-Account Access

Either:

- Add us as a user on your GHL sub-account (preferred), or
- Give us your GHL API key and Location ID (your consultant will tell you exactly where to find these)

We need access to your:

- **Contacts** — so the AI knows who's in your pipeline
- **Pipelines** — so we can move leads through stages automatically
- **Workflows** — so we can trigger actions based on lead score

2. Your Phone Number for Voice Calls

We'll set up a dedicated number (or use your existing one) for the AI to call leads from. It needs to be in Australian format.

If you want the AI to sound like a specific person from your team, we can clone their voice using ElevenLabs (takes about 10 minutes of audio).

3. Your Key Business Info

We need to programme the AI with:

- **Company name** and what you do
- **Service areas** (postcodes, states, or cities you cover)
- **System sizes and pricing** (approximate is fine — the AI won't quote exact prices, just ranges)
- **Current government rebates** and any special offers
- **What makes you different** from other solar companies in your area

This takes about 30 minutes in a Zoom call.

4. Your Lead Qualification Criteria

Tell us what a great lead looks like for you. For example:

- Homeowner (not renter)
- Monthly electricity bill over $250
- Roof less than 15 years old
- Not already got solar

The AI scores leads against these criteria. Anything scoring 7 or higher gets an automatic outbound call.

What Happens With Your Leads

Here's the journey a new lead takes through the system:

Step 1: Lead arrives

A new contact is created in GHL (from your website form, Facebook ad, or direct entry).

Step 2: AI scores the lead (< 5 seconds)

The AI reads the lead's details and gives them a score from 1 to 10:

- Score 7–10 → "Call now" (hot lead)
- Score 5–6 → "Nurture" (worth following up manually)
- Score 1–4 → "Disqualify" (unlikely to convert)

Step 3: Hot leads get an AI call

Within minutes of arriving, the AI calls your hot leads. It:

- Asks about their electricity bill, roof type, and home ownership
- Answers common questions about solar in plain language
- Books an assessment if they're interested
- Creates a GHL task for your sales team if they ask for a callback

Step 4: You see the result in GHL

After every call, the lead record in GHL is updated with:

- A summary of the conversation
- Tags like "assessment-booked" or "callback-requested"
- The recording URL so you can listen back

Step 5: You get a weekly report

Every Monday, you receive a performance summary showing call volume, booking rate, lead scores, and what the AI is learning.

What the AI Will and Won't Do

The AI will:

- Introduce itself clearly (it doesn't pretend to be human — though it won't volunteer that it's an AI unless asked directly)
- Collect the information your sales team needs before a site visit
- Book assessment appointments if you've connected your calendar
- Handle objections politely and professionally
- Transfer to a human if the prospect insists

The AI won't:

- Quote exact prices (it gives ranges and says your team will confirm)
- Pressure or rush prospects
- Call leads who are on the Do Not Call Register
- Make promises about savings that can't be guaranteed

Your Dashboard

You have access to a live dashboard showing:

- **Lead pipeline** — how many leads are at each stage
- **Recent calls** — outcome and score for every AI call
- **Weekly budget** — how much has been spent on experiments this week
- **AI learnings** — what's working and what isn't

Your consultant will send you the dashboard link and login credentials.

Approval Flow

Some things require your approval before the AI acts on them. You'll receive a Slack message (or email if Slack isn't set up) with an Approve or Reject button.

This typically happens for:

- New experiment ideas the AI wants to test
- Budget requests above the weekly threshold
- Any action that would spend money externally

You can approve or reject with a single button click. If you don't respond within 24 hours, the experiment stays on hold — nothing happens without your approval.

What You'll See in GHL

The system adds these tags to contacts automatically:

TAG	MEANING
`voice-ai-qualified`	AI called and qualified the lead
`assessment-booked`	Assessment appointment was booked on the call
`callback-requested`	Lead asked for a human callback
`hot-lead`	Score was 7 or higher
`battery-interest`	Lead mentioned interest in battery storage
`ev-owner`	Lead mentioned they own an electric vehicle
`not-interested-voice`	Lead declined on the call

You can use these tags to build GHL automations (for example, triggering a task for your sales team when `assessment-booked` is added).

Costs and What's Included in Your Retainer

Your monthly retainer covers:

ITEM	INCLUDED
Lead qualification scoring	All leads
AI outbound calls	Up to ~200 calls/month
Weekly performance reports	Yes
Dashboard access	Yes
GHL integration and maintenance	Yes
Support (email + Slack)	Business hours

API costs (OpenAI, Retell, ElevenLabs) run approximately $50–100 AUD per month at typical volumes and are included in your retainer. If your call volume significantly exceeds 200 calls per month, we'll discuss a small overage charge.

Privacy and Data

All lead data is stored in a secure database on your consultant's server. Data is not shared with any third parties except the API services used to run the system (OpenAI for AI, Retell for calls, GoHighLevel for your CRM).

Call recordings are stored by Retell and linked in your GHL contact record. You can request deletion of any recording at any time.

Getting Help

For urgent issues (system down, calls not working):

Contact your consultant directly via the number or email provided in your onboarding email.

For general questions or changes:

Email or Slack your consultant. Turnaround is typically same business day.

To pause the system:

Your consultant can pause all outbound calls or lead processing with one command. Just ask.

Checklist: Before We Go Live

Your consultant will work through this with you:

- [ ] GHL access confirmed

- [ ] Phone number set up and linked to the AI agent
- [ ] Company knowledge base written (products, pricing, service areas, FAQs)
- [ ] Lead qualification criteria agreed
- [ ] Test call done and voice approved
- [ ] GHL webhooks connected and tested
- [ ] Approval notifications set up (Slack or email)
- [ ] Dashboard access shared
- [ ] First weekly report template reviewed

Typical time from signed agreement to live system: **3–5 business days**.

Cost & Usage Tracking

Overview

Every API call made by the system — OpenAI, Retell, ElevenLabs — is logged to the `api_usage` table with a calculated cost in USD. This makes it possible to track per-client profitability, identify expensive operations, and project monthly costs.

File: `tracking/cost_tracker.py`

api_usage Table Schema

COLUMN	TYPE	DESCRIPTION
`id`	INTEGER PK	Auto-increment
`recorded_at`	TEXT	UTC timestamp
`service`	TEXT	`openai` / `retell` / `elevenlabs`
`operation`	TEXT	e.g. `chat_completion`, `voice_call`, `tts`
`model`	TEXT	e.g. `gpt-4o`, `retell-agent`, voice ID
`units`	REAL	Tokens, minutes, or characters consumed
`unit_type`	TEXT	`tokens` / `minutes` / `characters`
`cost_usd`	REAL	Estimated cost in USD (6 decimal places)
`call_id`	TEXT	Links to `call_logs.call_id`
`client_id`	TEXT	Links to `company_profiles`
`metadata`	TEXT	JSON with extra context (token split, agent_id, etc.)

Pricing Reference

Current rates used by the cost tracker (update in `tracking/cost_tracker.py` if rates change):

OpenAI

MODEL	INPUT (PER 1M TOKENS)	OUTPUT (PER 1M TOKENS)
`gpt-4o`	$2.50 USD	$10.00 USD
`gpt-4o-mini`	$0.15 USD	$0.60 USD
`gpt-4-turbo`	$10.00 USD	$30.00 USD
`gpt-3.5-turbo`	$0.50 USD	$1.50 USD

Retell AI

$0.07 USD per minute of call duration.

ElevenLabs

$0.005 USD per 1,000 characters of synthesised speech (Starter plan).

AUD conversion: The tracker uses a fixed rate of **1.55 AUD per USD**. Update `AUD_PER_USD` in `cost_tracker.py` if the rate drifts significantly.

Logging Functions

log_openai

```
from tracking.cost_tracker import log_openai

cost_usd = log_openai(
    model="gpt-4o",
    prompt_tokens=512,
    completion_tokens=128,
    call_id="call_abc123",     # optional
    client_id="sunpower_perth", # optional
    operation="chat_completion", # default
)
```

Called automatically by `voice/call_handler.py` on every GPT-4o LLM call during a conversation.

log_retell_call

```
from tracking.cost_tracker import log_retell_call

cost_usd = log_retell_call(
    call_id="call_abc123",
    duration_seconds=180,
    client_id="sunpower_perth",
    agent_id="agent_xyz",
)
```

Called by `voice/call_handler.py` in the post-call webhook handler.

log_elevenlabs

```
from tracking.cost_tracker import log_elevenlabs

cost_usd = log_elevenlabs(
    characters=1500,
    voice_id="11labs-Adrian",
    call_id="call_abc123",
    client_id="sunpower_perth",
    operation="tts",           # or "conversation"
)
```

Called when ElevenLabs TTS or Conversational AI is used.

Reporting API

get_cost_summary(days=7)

Returns total cost breakdown by service for the last N days.

```
from tracking.cost_tracker import get_cost_summary

summary = get_cost_summary(days=30)
# {
#   "period_days": 30,
#   "breakdown": {
#     "openai":     {"cost_usd": 12.50, "calls": 480, "units": 5000000, "unit_type": "tokens"},
#     "retell":     {"cost_usd": 8.40,  "calls": 120},
#     "elevenlabs": {"cost_usd": 0.75,  "calls": 120}
#   },
#   "total_usd": 21.65,
#   "total_aud": 33.56,
#   "generated_at": "2026-03-09T08:00:00"
# }
```

get_daily_costs(days=30)

Day-by-day totals for trend analysis.

```
daily = get_daily_costs(days=7)
# [
#   {"date": "2026-03-09", "cost_usd": 0.82, "cost_aud": 1.27, "api_calls": 24},
#   {"date": "2026-03-08", "cost_usd": 1.10, "cost_aud": 1.71, "api_calls": 31},
#   ...
# ]
```

get_call_cost(call_id)

Total cost for a single voice call across all services.

```
call_cost = get_call_cost("call_abc123")
# {
#   "call_id": "call_abc123",
#   "total_usd": 0.094,
#   "total_aud": 0.15,
#   "services": [
#     {"service": "openai",     "cost_usd": 0.024, "units": 2400,  "unit_type": "tokens"},
#     {"service": "retell",     "cost_usd": 0.063, "units": 0.9,   "unit_type": "minutes"},
#     {"service": "elevenlabs", "cost_usd": 0.007, "units": 1500,  "unit_type": "characters"}
#   ]
# }
```

get_client_costs(client_id, days=30)

Cost totals for a specific solar company client. Use this to check per-client profitability against their retainer.

```
client = get_client_costs("sunpower_perth", days=30)
# {
#   "client_id": "sunpower_perth",
#   "period_days": 30,
#   "total_usd": 14.20,
#   "total_aud": 22.01,
#   "breakdown": [
#     {"service": "openai",  "cost_usd": 8.50, "calls": 340},
#     {"service": "retell",  "cost_usd": 5.70, "calls": 82},
#   ]
# }
```

get_projected_monthly_cost()

Extrapolates the last 7 days of usage to a monthly estimate and calculates margin against the $1,500–2,000 AUD retainer.

```
projection = get_projected_monthly_cost()
# {
#   "last_7_days_usd": 5.20,
#   "projected_month_usd": 22.29,
#   "projected_month_aud": 34.55,
#   "margin_note": "At $35 AUD/month costs vs $1,500-2,000 retainer = ~98% margin"
# }
```

HTTP API

All cost reporting is available via the Human Gate API on port 5000.

```
# Overall 7-day summary
curl -H "Authorization: Bearer $GATE_API_KEY" \
  http://localhost:5000/costs

# 30-day view
curl -H "Authorization: Bearer $GATE_API_KEY" \
  "http://localhost:5000/costs?days=30"

# Cost for one call
curl -H "Authorization: Bearer $GATE_API_KEY" \
  "http://localhost:5000/costs?call_id=call_abc123"

# Cost for one client (30 days)
curl -H "Authorization: Bearer $GATE_API_KEY" \
  "http://localhost:5000/costs?client_id=sunpower_perth&days=30"
```

Typical Monthly Cost Estimates

At typical Australian solar SME volumes:

COMPONENT	VOLUME	MONTHLY COST (USD)
OpenAI GPT-4o	~2M tokens (agent runs + voice)	~$10–20
Retell AI	~100 calls × 3 min avg	~$21
ElevenLabs	~300,000 chars/month	~$1.50
Total		**~$32–42 USD (~$50–65 AUD)**

This leaves a margin of approximately 95–97% on a $1,500 AUD/month retainer.

Querying the Database Directly

For ad-hoc analysis, query the `api_usage` table directly:

```
sqlite3 swarm.db
```

```
-- Total cost by service this month
SELECT service,
       SUM(cost_usd) AS total_usd,
       COUNT(*) AS call_count
FROM api_usage
WHERE recorded_at >= date('now', 'start of month')
GROUP BY service
ORDER BY total_usd DESC;

-- Most expensive individual calls
SELECT call_id,
       SUM(cost_usd) AS total_usd,
       client_id,
       MIN(recorded_at) AS started_at
FROM api_usage
WHERE call_id IS NOT NULL
GROUP BY call_id
ORDER BY total_usd DESC
LIMIT 20;

-- Daily trend
SELECT DATE(recorded_at) AS day,
       ROUND(SUM(cost_usd), 4) AS cost_usd,
       COUNT(*) AS api_calls
FROM api_usage
WHERE recorded_at >= date('now', '-30 days')
GROUP BY day
ORDER BY day DESC;
```

Budget Alerts

The circuit breaker monitors experiment budget burn (not API costs). If budget spending exceeds 150% of the planned daily rate, the circuit breaker moves to Orange level.

To set up a budget alert for total API costs, add a check in your weekly retrospective:

```
from tracking.cost_tracker import get_cost_summary

summary = get_cost_summary(days=7)
if summary["total_aud"] > 50:  # Alert if weekly API costs exceed $50 AUD
    from notifications.slack_notifier import notify
    notify(f"Weekly API costs: ${summary['total_aud']} AUD — review usage")
```

Adding a New Tracked Service

1. Add a pricing entry to the `PRICING` dict in `tracking/cost_tracker.py`
2. Create a `log_()` function following the same pattern as `log_openai()`
3. Call `_insert()` with the correct `service`, `units`, `unit_type`, and `cost_usd`

4. Call your logging function wherever the service API is invoked

Troubleshooting Guide

Common errors, symptoms, and fixes.

Startup Errors

`ModuleNotFoundError: No module named 'openai'`

Cause: Dependencies not installed, or running Python outside the virtual environment.

Fix:

```
source .venv/bin/activate
pip install -r requirements.txt
```

`sqlite3.OperationalError: no such table: experiments`

Cause: Database not initialised.

Fix:

```
python3 -c "from memory.database import init_db; init_db()"
```

If the error persists after init, check that `DATABASE_PATH` in `.env` points to a writable location.

`Address already in use` (port 5000, 5001, 5002, or 5003)

Cause: A previous instance of `main.py` is still running, or another process is using the port.

Fix:

```
# Find the process
lsof -i :5000

# Kill it
kill -9 <PID>
```

Or change the port in `.env`:

```
PORT_HUMAN_GATE=5010
```

`ImportError: cannot import name 'flask_limiter'`

Cause: `flask-limiter` not installed.

Fix:

```
pip install flask-limiter
```

OpenAI / AI Errors

`openai.AuthenticationError: Incorrect API key`

Cause: `OPENAI_API_KEY` is missing, wrong, or has leading/trailing whitespace.

Fix:

1. Check `.env`: `OPENAI_API_KEY=sk-...` (no quotes, no trailing spaces)
2. Verify the key at platform.openai.com
3. Confirm the virtual environment loaded the `.env`: `python3 -c "import config; print(config.OPENAI_API_KEY[:8])"`

`openai.RateLimitError`

Cause: Too many requests per minute, or API tier limit reached.

Fix:

- Check your OpenAI usage tier at platform.openai.com/usage
- The system's try/except wrappers log and continue — individual agent runs may produce partial results
- Upgrade to a higher OpenAI tier if this is recurring

`openai.BadRequestError: maximum context length exceeded`

Cause: A system prompt or transcript is too long for the model's context window.

Fix:

- For voice calls: reduce the number of `company_knowledge` items in `voice/prompt_templates.py`
- For the General: the `_generate_ideas()` function will fall back to `_mock_ideas()` and log an error

---

The General generates mock ideas instead of real ones

Cause: OpenAI API key is not configured, or the API call failed.

Symptom: Output contains "Solar Prospecting Automation" and "Solar ROI Content Engine" — these are the mock idea names.

Fix:

1. Verify `OPENAI_API_KEY` is set
2. Check the logs for the preceding error: `[GENERAL] GPT-4o ideas failed:...`
3. Confirm the key has sufficient credits

GoHighLevel / CRM Errors

`[GHL] Not configured — missing GHL_API_KEY or GHL_LOCATION_ID`

Cause: GHL credentials not set.

Fix:

Add to `.env`:

```
GHL_API_KEY=your-key
GHL_LOCATION_ID=your-location-id
```

The system runs in mock mode without these — all CRM calls are silently skipped.

`[GHL WEBHOOK] new-lead: invalid signature` **— leads not being processed**

Cause: `GHL_WEBHOOK_SECRET` in `.env` doesn't match the signing secret in GHL, or the secret isn't set.

Fix:

1. Go to GHL → Settings → Integrations → Webhooks
2. Find your webhook and copy the Signing Secret
3. Add to `.env`: `GHL_WEBHOOK_SECRET=your-signing-secret`
4. Restart `main.py`

If you don't have a signing secret configured in GHL, remove `GHL_WEBHOOK_SECRET` from `.env` — the system will skip validation and allow all requests (acceptable for trusted networks only).

`crm_router: active CRM is 'none'`

Cause: No CRM credentials are configured.

Fix: Set at least one of: `GHL_API_KEY` + `GHL_LOCATION_ID`, `HUBSPOT_API_KEY`, or Salesforce credentials. The router selects the first configured CRM in priority order: GHL → HubSpot → Salesforce.

Leads arrive in the database but don't appear in GHL

Cause: The system stores leads locally first, then syncs to GHL. Sync fails silently if GHL is not configured or the API call fails.

Fix:

1. Check that `GHL_API_KEY` and `GHL_LOCATION_ID` are set
2. Check the logs for `[GHL CLIENT]` error lines
3. Test the GHL connection: `curl http://localhost:5003/api/crm/status`

Voice AI Errors

Outbound calls not triggering for high-score leads

Cause: `RETELL_API_KEY` not configured, or the lead is missing a phone number.

Fix:

1. Check `RETELL_API_KEY` in `.env`
2. Verify the lead has a `phone` field: `sqlite3 swarm.db "SELECT phone FROM leads ORDER BY id DESC LIMIT 5;"`
3. Check qualification agent logs for: `[QUALIFICATION] Skipping outbound call — no phone`

`/voice/response` returns 500 errors during calls

Cause: GPT-4o call failed mid-conversation. This causes Retell to hear silence or fall back to its own LLM.

Fix:

1. Check logs for `[VOICE] GPT-4o call failed:`
2. Confirm OpenAI rate limits aren't being hit (voice calls make 8–20 GPT-4o calls each)
3. Check that the `_call_contexts` dict has a valid entry for the `call_id` — if the server restarted mid-call, context is lost

---

Voice agent says "I'm sorry, I'm having trouble right now"

Cause: The system prompt or transcript exceeded GPT-4o's context limit, or the API call timed out.

Fix:

- Reduce the `company_knowledge` payload in `voice/prompt_templates.py`
- Check the OpenAI timeout setting in `voice/call_handler.py`

ElevenLabs voice not working, falling back to default

Cause: `ELEVENLABS_API_KEY` not set, or the `elevenlabs_voice_id` on the company profile is invalid.

Fix:

1. Set `ELEVENLABS_API_KEY` in `.env`
2. Verify the voice ID in the ElevenLabs dashboard
3. Check logs for `[VOICE] ElevenLabs error:`

Circuit Breaker

System halted — Red circuit breaker

Symptom: The General stops running. New experiments are not generated. Logs show `[CIRCUIT BREAKER] System halted — Red level`.

Cause: Either 5+ consecutive experiment failures, or a single experiment lost more than 40% of the weekly budget.

Fix: Requires explicit human reset:

```
curl -X POST http://localhost:5000/approve-breaker \
  -H "Authorization: Bearer $GATE_API_KEY" \
  -H "Content-Type: application/json" \
  -d '{"approved_by": "martin"}'
```

Before resetting, review what triggered it:

```
sqlite3 swarm.db "SELECT * FROM circuit_breaker_log ORDER BY triggered_at DESC LIMIT 5;"
```

Experiments keep failing and not learning from history

Cause: Warm memory files may be missing or corrupt.

Fix:

```
ls memory/knowledge/
# Should contain: experiments.json, learnings.json, verticals.json
```

If files are missing:

```
mkdir -p memory/knowledge
echo '[]' > memory/knowledge/experiments.json
echo '[]' > memory/knowledge/learnings.json
echo '{}' > memory/knowledge/verticals.json
```

Scheduler / Timing Issues

Agents not running on schedule

Cause: APScheduler failed to start, or a previous job run errored and blocked the thread.

Fix:

1. Check logs at startup for `[SCHEDULER]` lines
2. Look for unhandled exceptions in previous job runs — APScheduler swallows them silently unless logging is configured
3. Restart `main.py` — APScheduler starts fresh on restart

To trigger an agent manually:

```
python3 cli.py run-general
python3 cli.py run-scout
```

Data collection not running

Symptom: `collected_data` table is empty or not growing.

Fix:

1. Check that at least one row exists in `collection_sources` with `active = 1`
2. Check logs for `[DATA COLLECTION]` errors
3. Run manually: `python3 -c "from data_collection.orchestrator import run; run()"`

Swarm Board Errors

Board shows "API Offline" or Overview tab is blank

Cause: The Dashboard API on port 5003 is not running, or CORS is blocking requests.

Fix:

1. Confirm `main.py` is running and the dashboard API started: `curl http://localhost:5003/api/health`
2. If running on a different port, update `VITE_API_URL` in `swarm-board/.env.local`
3. If the frontend is on a non-localhost domain, set `FRONTEND_URL` in `.env` and restart `main.py`

Board tasks reset to seed tasks on every reload

Cause: The browser's localStorage was cleared, or the board is running in private/incognito mode.

Fix: Use a regular (non-incognito) browser window. Tasks persist in `localStorage` under the key `swarm_tasks`.

---

npm error: `Cannot find module @vitejs/plugin-react`

Cause: Node modules not installed.

Fix:

```
cd swarm-board
npm install
```

Slack Errors

Slack notifications not sending

Cause: `SLACK_WEBHOOK_URL` not configured, or the webhook URL has been revoked.

Fix:

1. Check `.env`: `SLACK_WEBHOOK_URL=https://hooks.slack.com/services/...`
2. Test the webhook: `curl -X POST -H 'Content-type: application/json' --data '{"text":"Test"}' $SLACK_WEBHOOK_URL`
3. If revoked, create a new incoming webhook at api.slack.com/apps

---

`/slack/actions` returning 403

Cause: Slack signature verification failed. `SLACK_SIGNING_SECRET` is wrong, or the request is too old (>5 minutes).

Fix:

1. Go to api.slack.com/apps → your app → Basic Information → App Credentials → Signing Secret
2. Update `SLACK_SIGNING_SECRET` in `.env`
3. Restart `main.py`

Database Errors

`sqlite3.OperationalError: database is locked`

Cause: Two processes are writing to the database simultaneously and WAL mode isn't resolving the contention fast enough.

Fix:

- The database uses WAL journal mode which handles concurrent access — this should be rare
- If it persists, check for zombie processes: `ps aux | grep python`
- Ensure only one instance of `main.py` is running
- If a backup tool is locking the file, exclude `swarm.db` from live backup

---

Column missing errors after upgrade (`no such column`)

Cause: A new column was added to the schema but the migration didn't run.

Fix: Migrations run automatically on every startup via `_apply_migrations()`. Simply restart `main.py`:

```
python3 main.py
```

If the error persists, check `memory/database.py` → `_apply_migrations()` to see if the column is in the migrations list. If not, add it and restart.

Getting More Debug Information

Enable verbose logging by adding to `.env`:

```
LOG_LEVEL=DEBUG
```

Or at runtime:

```
import logging
logging.basicConfig(level=logging.DEBUG)
```

To watch the database in real time during a run:

```
watch -n 2 'sqlite3 swarm.db "SELECT status, COUNT(*) FROM experiments GROUP BY status;"'
```